

KhataAI

REST API Design Reference

Section 4 — Complete Edition v4.0

20 Modules • 70+ Endpoints • All New: Suppliers, Purchases, Payments, Reports, Categories, Feature Flags

Base URL: /api/v1 • Auth: Bearer JWT • Format: JSON • FastAPI + MongoDB

Global API Standards

• All endpoints versioned under **/api/v1** • Content-Type: **application/json** (multipart for files) • Authorization: **Bearer JWT** on all non-public endpoints • Response envelope: **success, message, data, errors** • List APIs support: **page, size, sort, search, filters** • Important actions produce **audit log** entries • **Never hard-delete** Customers, Products, Users, Transactions — use soft delete / status / reversal • Backend always fetches price from DB — **never trust client-submitted prices** • CGST + SGST for intra-state supply • IGST for inter-state • AI extracts data; backend calculates all financials

Table of Contents

4.1 Authentication

POST /auth/send-otp – Send OTP to user's phone number.
POST /auth/verify-otp – Verify OTP and issue JWT + Refresh tokens.
POST /auth/refresh – Generate new access token from refresh token.
POST /auth/logout – Logout current session.
GET /auth/me – Return currently authenticated user profile.
PUT /auth/change-password – Change authenticated user password.
POST /auth/forgot-password – Send password reset link to email. [Future]
POST /auth/reset-password – Reset password using token. [Future]

4.2 Users & Businesses

GET /users/me – Get logged-in user's full profile.
PUT /users/me – Update user profile (name, language, timezone).
POST /businesses – Create a new business/store.
GET /businesses – List all businesses.
GET /businesses/{id} – Get business details.
PUT /businesses/{id} – Update business info.
DELETE /businesses/{id} – Soft-delete a business.

4.3 Store & Settings

GET /store/profile – Return current store/business profile.
PUT /store/profile – Update business information.
POST /store/logo – Upload store logo (multipart/form-data).
GET /store/settings – Return store configuration.
PUT /store/settings – Update invoice numbering and store settings.
GET /settings/features – Get enabled feature flags for this store.
PUT /settings/features – Enable or disable feature modules.

4.4 Categories

POST /categories – Create a new product category.
GET /categories – List all categories.
PUT /categories/{id} – Update category name or description.
PATCH /categories/{id}/status – Enable or disable a category (soft).

4.5 Customers

POST /customers – Create a new customer.
GET /customers – List customers with pagination and search.
GET /customers/{id} – Get customer details.
PUT /customers/{id} – Update customer information.
PATCH /customers/{id}/status – Change customer status. Never delete.
GET /customers/{id}/statement – Get customer ledger/transaction history.

4.6 Products & Inventory

POST /products – Create a new product.
GET /products – List all products.
GET /products/search – Fast product search by name, SKU, barcode, alias.
GET /products/{id} – Get product details.
PUT /products/{id} – Update product details.
PATCH /products/{id}/status – Enable or disable a product (soft).
GET /inventory – Get current stock levels for all products.
POST /inventory/stock-in – Record purchase / received stock.
POST /inventory/stock-adjustment – Manual stock correction (damage, loss, etc.).
GET /inventory/history – Get full inventory movement history.

4.7 Suppliers

POST /suppliers – Create a new supplier.
GET /suppliers – List all suppliers.
GET /suppliers/{id} – Get supplier details.
PUT /suppliers/{id} – Update supplier information.
PATCH /suppliers/{id}/status – Enable or disable a supplier (soft).
GET /suppliers/{id}/statement – Get supplier ledger / payment history.

4.8 Purchase Module

POST /purchases – Create a new purchase bill from supplier.
GET /purchases – List all purchase bills.
GET /purchases/{id} – Get purchase bill details.
POST /purchases/{id}/receive – Mark purchase as received and update stock.
POST /purchases/{id}/cancel – Cancel a purchase bill.

4.9 Invoices & Transactions

POST /invoices – Create a new sales invoice.
GET /invoices – List invoices with filters.
GET /invoices/{id} – Get invoice details.
GET /invoices/{id}/pdf – Download invoice as PDF.
POST /invoices/{id}/cancel – Cancel an invoice.
POST /transactions/manual-bill – Create transaction/invoice manually (simplified flow).
POST /transactions/voice-bill – Create transaction from pre-processed voice audio ID.
POST /transactions/settle – Record payment for an unpaid/partial transaction.
GET /transactions/history – Get transaction history.
GET /transactions/{id} – Get transaction details.
POST /transactions/{id}/reverse – Reverse a transaction (refund/void). Never delete.

4.10 Payments

POST /payments – Record a payment against a transaction or advance.
GET /payments – List all payment records.
GET /payments/{id} – Get payment details.

4.11 Voice Processing

POST /voice/process – Upload audio and create a transaction draft.
POST /voice/confirm – Confirm a voice draft and create the final transaction.
GET /voice/history – Get history of voice processing sessions.
DELETE /voice/history/{id} – Delete a voice draft record (failed/test drafts only).

4.12 Expenses

POST /expenses – Create a new expense record.
GET /expenses – List all expenses.
GET /expenses/{id} – Get expense details.

4.13 Reports & Export

GET /reports/sales – Sales report with totals, GST breakdown, top products.
GET /reports/expenses – Expense report grouped by category.
GET /reports/gst – GST report: CGST, SGST, IGST summary for filing.
GET /reports/profit-loss – Profit & Loss: Sales minus Expenses minus Purchase Cost.
GET /reports/export – Export any report as CSV or Excel.

4.14 AI Assistant

POST /ai/chat – Ask the business AI assistant a natural language question.
GET /ai/history – Retrieve previous AI conversations.
DELETE /ai/history/{id} – Delete a specific AI conversation thread.

4.15 Analytics & Dashboard

GET /dashboard – Dashboard summary for today.
GET /analytics/top-products – Best selling products by qty or revenue.
GET /analytics/top-customers – Top customers by revenue.
GET /analytics/monthly-sales – Monthly sales trend (12-month rolling).

4.16 Files & Search

POST /files/upload – Upload a file (invoice PDF, receipt image, audio).
GET /files/{id} – Download an uploaded file.
DELETE /files/{id} – Delete an uploaded file.
GET /search – Global search across customers, invoices, products, suppliers.

4.17 Notifications & Audit

GET /notifications – Fetch all notifications (unread first).
PATCH /notifications/{id}/read – Mark a notification as read.
GET /audit-logs – Get audit log history for this tenant.

4.18 OCR — Future Feature

POST /ocr/process-invoice – Extract line items from a scanned invoice or PDF.

4.19 Admin

GET /admin/users – List all platform users.
GET /admin/businesses – List all businesses on the platform.
GET /admin/reports – Platform-level aggregate reports.

4.20 Health Check

GET /health – Service health check endpoint.

4. Authentication

1

Phone OTP → JWT access + refresh token flow

POST /auth/send-otpAuth: **Public****Purpose:** Send OTP to user's phone number.**REQUEST**

```
{"phone": "+919876543210"}
```

RESPONSE

```
{"success": true, "message": "OTP sent"}
```

VALIDATION & PROCESSING

- Validate phone number format (+91 prefix)
- Rate-limit: 5 requests/hour per phone
- Generate 6-digit OTP
- Store in Redis (TTL 5 min)
- Send SMS asynchronously

POST /auth/verify-otpAuth: **Public****Purpose:** Verify OTP and issue JWT + Refresh tokens.**REQUEST**

```
{"phone": "+919876543210", "otp": "483912"}
```

RESPONSE

```
{"access_token": "jwt", "refresh_token": "jwt", "expires_in": 3600, "user": {"id": "uuid", "name": "John"}}
```

VALIDATION & PROCESSING

- Validate OTP from Redis
- Auto-create user on first login
- Issue JWT + Refresh Token
- Delete OTP from Redis after success

POST /auth/refreshAuth: **Refresh Token****Purpose:** Generate new access token from refresh token.**REQUEST**

```
{"refresh_token": "xxxxx"}
```

RESPONSE

```
{"access_token": "new_jwt"}
```

VALIDATION & PROCESSING

- Validate refresh token signature
- Revoke token if expired or blacklisted

POST /auth/logoutAuth: **JWT****Purpose:** Logout current session.**REQUEST**

```
{}
```

RESPONSE

```
{"success": true, "message": "Logged out"}
```

VALIDATION & PROCESSING

- Blacklist active JWT in Redis
- Remove refresh token
- Invalidate session

GET /auth/meAuth: **JWT****Purpose:** Return currently authenticated user profile.**REQUEST**

```
—
```

RESPONSE

```
{"id": "uuid", "name": "John", "phone": "+91...", "role": "OWNER"}
```

VALIDATION & PROCESSING

- User must exist and be ACTIVE

PUT /auth/change-passwordAuth: **JWT****Purpose:** Change authenticated user password.**REQUEST**

```
{"old_password": "Old@123", "new_password": "New@123"}
```

RESPONSE

```
{"success": true}
```

VALIDATION & PROCESSING

- Verify old password hash
- Validate new password strength
- Update with bcrypt (rounds ≥ 12)

POST /auth/forgot-passwordAuth: **Public****Purpose:** Send password reset link to email. [Future]**REQUEST**

```
{"email": "user@example.com"}
```

RESPONSE

```
{"message": "Reset link sent"}
```

VALIDATION & PROCESSING

- Validate email exists
- Generate secure token
- Queue reset email

POST /auth/reset-password

Auth: Public

Purpose: Reset password using token. [Future]

REQUEST

```
{ "token": "abc", "password": "New@123" }
```

RESPONSE

```
{ "success": true }
```

VALIDATION & PROCESSING

- Validate token expiry
- Update password
- Invalidate token after use

4. Users & Businesses

2 User profile and multi-tenant business management

GET /users/me

Auth: JWT

Purpose: Get logged-in user's full profile.

REQUEST

–

RESPONSE

```
{ "id": "uuid", "name": "John", "phone": "+91...", "role": "OWNER" }
```

VALIDATION & PROCESSING

- User must exist and be ACTIVE

PUT /users/me

Auth: JWT

Purpose: Update user profile (name, language, timezone).

REQUEST

```
{ "name": "John Doe", "language": "en", "timezone": "Asia/Kolkata" }
```

RESPONSE

```
{ "success": true }
```

VALIDATION & PROCESSING

- Validate all fields
- Sanitise inputs

POST /businesses

Auth: JWT

Purpose: Create a new business/store.

REQUEST

```
{ "name": "ABC Store", "gst_number": "24ABCDE1234A1Z5", "currency": "INR" }
```

RESPONSE

```
{ "business_id": "uuid" }
```

VALIDATION & PROCESSING

- Validate GST number format
- Ensure business name uniqueness
- Assign owner role

GET /businesses

Auth: JWT

Purpose: List all businesses.

REQUEST

```
?page=1&size;=20&search;=abc
```

RESPONSE

```
{ "page": 1, "size": 20, "total": 1, "items": [...] }
```

VALIDATION & PROCESSING

- Pagination, sorting, filtering
- Tenant-scoped results

GET /businesses/{id}

Auth: JWT

Purpose: Get business details.

REQUEST

–

RESPONSE

```
{ "id": "uuid", "name": "ABC Store" }
```

VALIDATION & PROCESSING

- Ownership check

PUT /businesses/{id}

Auth: JWT

Purpose: Update business info.

REQUEST

```
{ "name": "ABC Pvt Ltd", "address": "Rajkot" }
```

RESPONSE

```
{ "success": true }
```

VALIDATION & PROCESSING

- Ownership check
- Validate fields

DELETE	/businesses/{id}	Auth: JWT
--------	------------------	-----------

Purpose: Soft-delete a business.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"success":true}</pre>	• Set deleted_at timestamp • Never hard delete

4.

3

Store & Settings

Business profile, logo, invoice config and feature flags

GET	/store/profile	Auth: JWT
-----	----------------	-----------

Purpose: Return current store/business profile.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"name":"ABC Store","gstn":"24ABCDE...", "upi":"abc@upi"}</pre>	• Return tenant-scoped store profile

PUT	/store/profile	Auth: JWT
-----	----------------	-----------

Purpose: Update business information.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"name":"ABC Store","address":"Rajkot","gstn":"24ABCDE..."}</pre>	<pre>{"success":true}</pre>	• Validate GST format • Ownership check

POST	/store/logo	Auth: JWT
------	-------------	-----------

Purpose: Upload store logo (multipart/form-data).

REQUEST	RESPONSE	VALIDATION & PROCESSING
multipart/form-data (JPEG/PNG)	<pre>{"url":"https://cdn/logo.png"}</pre>	• Validate file type (JPEG/PNG) • Max size 2MB • Upload to S3/CDN

GET	/store/settings	Auth: JWT
-----	-----------------	-----------

Purpose: Return store configuration.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"invoice_prefix":"INV","currency":"INR","financial_year":"2026-27"}</pre>	• Return tenant-scoped config

PUT	/store/settings	Auth: JWT
-----	-----------------	-----------

Purpose: Update invoice numbering and store settings.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"invoice_prefix":"INV","currency":"INR"}</pre>	<pre>{"success":true}</pre>	• Validate prefix format • Validate currency code

GET	/settings/features	Auth: JWT
-----	--------------------	-----------

Purpose: Get enabled feature flags for this store.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"gst":true,"voice":true,"ocr":false,"suppliers":true,"inventory":true,"ai":true}</pre>	• Return tenant-scoped feature toggles

PUT

/settings/features

Auth: JWT

Purpose: Enable or disable feature modules.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"gst":true,"suppliers":true,"ocr":true,"voice":true}</pre>	<pre>{"success":true}</pre>	• Validate feature keys • Disabling GST hides HSN/tax fields in UI • Disabling suppliers hides purchase module • Audit log entry

4.4

Categories

Product category management

POST

/categories

Auth: JWT

Purpose: Create a new product category.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"name":"Fabrics","description":"Textile products"}</pre>	<pre>{"category_id":"uuid"}</pre>	• Validate name uniqueness within store • Sanitise name

GET

/categories

Auth: JWT

Purpose: List all categories.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?page=1&search=fab</pre>	<pre>{"items":[{"id":"uuid","name":"Fabrics","product_count":42},...]}</pre>	• Pagination • Tenant-scoped • Include product_count per category

PUT

/categories/{id}

Auth: JWT

Purpose: Update category name or description.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"name":"Textiles & Fabrics"}</pre>	<pre>{"success":true}</pre>	• Ownership check • Validate name uniqueness

PATCH

/categories/{id}/status

Auth: JWT

Purpose: Enable or disable a category (soft).

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"active":false}</pre>	<pre>{"success":true}</pre>	• Products in disabled category remain accessible • Audit log

4.5

Customers

Customer directory — never hard delete

POST

/customers

Auth: JWT

Purpose: Create a new customer.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"name":"Rahul","phone":"9999999999","email":"a@gmail.com"}</pre>	<pre>{"customer_id":"uuid"}</pre>	• Phone uniqueness within business • Validate email format • Sanitise name

GET

/customers

Auth: JWT

Purpose: List customers with pagination and search.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?page=1&search;=rahul&type;=WHOLES ALE	<pre>{ "items": [...], "total": 50, "page": 1 }</pre>	• Pagination, search, filter by type • Tenant-scoped results

GET

/customers/{id}

Auth: JWT

Purpose: Get customer details.

REQUEST	RESPONSE	VALIDATION & PROCESSING
–	<pre>{ "id": "uuid", "name": "Rahul", "phone": "9999999999", "outstanding_balance": 4500 }</pre>	• Ownership check

PUT

/customers/{id}

Auth: JWT

Purpose: Update customer information.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "name": "Rahul Patel", "credit_limit": 50000 }</pre>	<pre>{ "success": true }</pre>	• Validate fields • Ownership check

PATCH

/customers/{id}/status

Auth: JWT

Purpose: Change customer status. Never delete.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "status": "INACTIVE" }</pre>	<pre>{ "success": true }</pre>	• Valid statuses: ACTIVE, INACTIVE, BLOCKED • Audit log entry

GET

/customers/{id}/statement

Auth: JWT

Purpose: Get customer ledger/transaction history.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?from=2026-04-01&to;=2026-06-30	<pre>{ "transactions": [...], "balance": 1500, "total_purchased": 95000 }</pre>	• Ownership check • Return full ledger with date filter

4.6

Products & Inventory

Product catalogue with HSN/GST and stock management

POST

/products

Auth: JWT

Purpose: Create a new product.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "name": "Coke", "selling_price": 40, "unit": "Piece", "gst_rate": 12 }</pre>	<pre>{ "product_id": "uuid", "sku": "PRD-001" }</pre>	• Validate name and price • Auto-generate SKU if not provided • Default status ACTIVE

GET

/products

Auth: JWT

Purpose: List all products.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?page=1&category;=Fabrics&status;=ACTIVE	<pre>{ "items": [...], "total": 100 }</pre>	• Pagination, category filter • Tenant-scoped

GET /products/searchAuth: **JWT****Purpose:** Fast product search by name, SKU, barcode, alias.

REQUEST

`?q=coke`

RESPONSE

`{"items": [...]}`

VALIDATION & PROCESSING

- Full-text + alias search
- Return ACTIVE products only by default
- Supports barcode scan query

GET /products/{id}Auth: **JWT****Purpose:** Get product details.

REQUEST

`—`

RESPONSE

`{"id":"uuid","name":"Coke","selling_price":40,"stock":200,"hsn_code":"2202"}`

VALIDATION & PROCESSING

- Ownership check

PUT /products/{id}Auth: **JWT****Purpose:** Update product details.

REQUEST

`{"selling_price":45,"hsn_code":"2202","gst_rate":12,"min_stock":20}`

RESPONSE

`{"success":true}`

VALIDATION & PROCESSING

- Validate fields
- Ownership check
- Audit log for price changes

PATCH

/products/{id}/status

Auth: **JWT****Purpose:** Enable or disable a product (soft).

REQUEST

`{"active":false}`

RESPONSE

`{"success":true}`

VALIDATION & PROCESSING

- Set active flag
- Audit log

GET /inventoryAuth: **JWT****Purpose:** Get current stock levels for all products.

REQUEST

`?low_stock=true`

RESPONSE

`{"items":[{"product_id":"uuid","name":"Coke","stock":5,"min_stock":20},...]}`

VALIDATION & PROCESSING

- low_stock=true returns only below-threshold products
- Tenant-scoped

POST /inventory/stock-inAuth: **JWT****Purpose:** Record purchase / received stock.

REQUEST

`{"product_id":"uuid","qty":20,"note":"Purchase from supplier","supplier_id":"uuid"}`

RESPONSE

`{"success":true}`

VALIDATION & PROCESSING

- Validate product exists
- Increment stock atomically (\$inc)
- Create inventory_history entry

POST /inventory/stock-adjustmentAuth: **JWT****Purpose:** Manual stock correction (damage, loss, etc.).

REQUEST

`{"product_id":"uuid","qty":-2,"reason":"Damaged"}`

RESPONSE

`{"success":true}`

VALIDATION & PROCESSING

- Validate product exists
- Adjust stock (positive or negative)
- Require reason for negative adjustments
- Create inventory_history entry

GET

/inventory/history

Auth: JWT

Purpose: Get full inventory movement history.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?product_id=uuid&type;=STOCK_OUT&page;=1</pre>	<pre>{"history":[...]}</pre>	• Pagination, filter by product/type • Tenant-scoped

4.7

Suppliers

Supplier / vendor directory and AP ledger [Business Mode]

POST

/suppliers

Auth: JWT

Purpose: Create a new supplier.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"name":"Ravi Textiles","phone":"9876541230","gstn":"24ZZAAA1234A1Z5"}</pre>	<pre>{"supplier_id":"uuid"}</pre>	• Phone uniqueness within store • Validate GSTIN format if provided • Default status ACTIVE

GET

/suppliers

Auth: JWT

Purpose: List all suppliers.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?page=1&search;=ravi</pre>	<pre>{"items":[{"id":"uuid","name":"Ravi Textiles","outstanding_balance":45000},...]}</pre>	• Pagination, search • Tenant-scoped • Requires suppliers feature enabled

GET

/suppliers/{id}

Auth: JWT

Purpose: Get supplier details.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>-</pre>	<pre>{"id":"uuid","name":"Ravi Textiles","phone":"...", "outstanding_balance":45000}</pre>	• Ownership check

PUT

/suppliers/{id}

Auth: JWT

Purpose: Update supplier information.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"payment_terms":"Net 30","bank_ifsc":"SBIN0001234"}</pre>	<pre>{"success":true}</pre>	• Validate fields • Ownership check

PATCH

/suppliers/{id}/status

Auth: JWT

Purpose: Enable or disable a supplier (soft).

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"status":"INACTIVE"}</pre>	<pre>{"success":true}</pre>	• Valid statuses: ACTIVE, INACTIVE • Audit log

GET

/suppliers/{id}/statement

Auth: JWT

Purpose: Get supplier ledger / payment history.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?from=2026-04-01&to;=2026-06-30</pre>	<pre>{"purchases":[...], "payments":[...], "outstanding_balance":45000}</pre>	• Ownership check • Return full payable ledger with date filter

4.

8

Purchase Module

Purchase bills and stock receiving [Business Mode]

POST

/purchases

Auth: JWT

Purpose: Create a new purchase bill from supplier.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"supplier_id":"uuid","items":[{"product_id":"uuid","qty":100,"rate":36,"gst_rate":5}], "invoice_ref":"SUP/2026/001"}</pre>	<pre>{"purchase_id":"uuid","status":"PENDING"}</pre>	• Validate supplier exists • Validate product stock capacity • Calculate CGST/SGST/IGST • Create purchase record — stock NOT updated yet (awaits receive step)

GET

/purchases

Auth: JWT

Purpose: List all purchase bills.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?status=PENDING&supplier_id=uuid&page=1</pre>	<pre>{"items":[...],"total":15}</pre>	• Pagination, filter by status/supplier/date • Tenant-scoped

GET

/purchases/{id}

Auth: JWT

Purpose: Get purchase bill details.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>-</pre>	<pre>{"id":"uuid","supplier":"Ravi Textiles","grand_total":45000,"status":"RECEIVED"}</pre>	• Ownership check

POST

/purchases/{id}/receive

Auth: JWT

Purpose: Mark purchase as received and update stock.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"received_at":"2026-06-25","notes":"All items received"}</pre>	<pre>{"success":true}</pre>	• Validate purchase is in PENDING/PARTIAL status • Increment product stock atomically • Create inventory_history entries for each item • Update supplier outstanding_balance • Set status to RECEIVED • Audit log

POST

/purchases/{id}/cancel

Auth: JWT

Purpose: Cancel a purchase bill.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"reason":"Supplier unable to deliver"}</pre>	<pre>{"success":true}</pre>	• Validate purchase not already RECEIVED • Require reason • Set status CANCELLED • Audit log

4.

9

Invoices & Transactions

Immutable sales ledger — never delete, use reversal

POST /invoices

Auth: **JWT**

Purpose: Create a new sales invoice.

REQUEST

```
{
  "customer_id": "uuid",
  "items": [
    {
      "product_id": "uuid",
      "qty": 2,
      "discount_percent": 5
    }
  ],
  "payment_mode": "UPI",
  "paid_amount": 200
}
```

RESPONSE

```
{
  "invoice_id": "uuid",
  "invoice_number": "SRD/2026-27/001",
  "status": "PARTIAL"
}
```

VALIDATION & PROCESSING

- Validate stock availability
- Fetch prices from DB (never trust client)
- Calculate GST (CGST+SGST or IGST)
- Deduct stock atomically
- Update customer outstanding_balance
- Generate PDF (async)
- Queue WhatsApp/email

GET /invoices

Auth: **JWT**

Purpose: List invoices with filters.

REQUEST

```
?status=PAID&from;=2026-04-01&page;=1
```

RESPONSE

```
{
  "items": [...],
  "total": 30
}
```

VALIDATION & PROCESSING

- Filters: status, date range, customer
- Pagination
- Tenant-scoped

GET /invoices/{id}

Auth: **JWT**

Purpose: Get invoice details.

REQUEST

–

RESPONSE

```
{
  "id": "uuid",
  "invoice_number": "SRD/001",
  "grand_total": 19422,
  "status": "PAID"
}
```

VALIDATION & PROCESSING

- Ownership check

GET /invoices/{id}/pdf

Auth: **JWT**

Purpose: Download invoice as PDF.

REQUEST

–

RESPONSE

Binary PDF response (stream from S3)

VALIDATION & PROCESSING

- Authorization check
- Return pre-generated PDF or generate on-demand

POST /invoices/{id}/cancel

Auth: **JWT**

Purpose: Cancel an invoice.

REQUEST

```
{
  "reason": "Customer requested cancellation"
}
```

RESPONSE

```
{
  "success": true
}
```

VALIDATION & PROCESSING

- Cannot cancel PAID invoice
- Restore stock on cancellation
- Update customer balance
- Audit log

POST /transactions/manual-bill

Auth: **JWT**

Purpose: Create transaction/invoice manually (simplified flow).

REQUEST

```
{
  "customer_id": "uuid",
  "items": [...]
}
```

RESPONSE

```
{
  "transaction_id": "uuid"
}
```

VALIDATION & PROCESSING

- Generate invoice
- Update stock
- Record in ledger

POST /transactions/voice-bill

Auth: **JWT**

Purpose: Create transaction from pre-processed voice audio ID.

REQUEST

```
{
  "audio_id": "uuid"
}
```

RESPONSE

```
{
  "draft_transaction": "uuid"
}
```

VALIDATION & PROCESSING

- Validate audio_id exists
- Create draft (use Voice module for full flow)

POST

/transactions/settle

Auth: JWT

Purpose: Record payment for an unpaid/partial transaction.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "transaction_id": "uuid", "mode": "UPI", "amount": 4422 }</pre>	<pre>{ "success": true, "balance_due": 0 }</pre>	• Valid modes: UPI, CASH, CARD, BANK_TRANSFER, CHEQUE, CREDIT • Update transaction status to PAID/PARTIAL • Update customer outstanding_balance • Audit log

GET

/transactions/history

Auth: JWT

Purpose: Get transaction history.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?page=1&customer_id=uuid&from;=2026-04-01</pre>	<pre>{ "items": [...] }</pre>	• Pagination, date filter • Tenant-scoped

GET

/transactions/{id}

Auth: JWT

Purpose: Get transaction details.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>—</pre>	<pre>{ "id": "uuid", "grand_total": 500, "status": "PAID", "items": [...] }</pre>	• Ownership check

POST

/transactions/{id}/reverse

Auth: JWT

Purpose: Reverse a transaction (refund/void). Never delete.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "reason": "Refund requested by customer" }</pre>	<pre>{ "success": true }</pre>	• Require reason • Create reversal entry • Restore stock • Update customer balance • Audit log

4.10

Payments

Standalone payment records for partial and advance payments

POST

/payments

Auth: JWT

Purpose: Record a payment against a transaction or advance.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "customer_id": "uuid", "transaction_id": "uuid", "amount": 5000, "mode": "UPI", "reference": "UTR123456" }</pre>	<pre>{ "payment_id": "uuid" }</pre>	• Validate transaction exists (if linked) • Amount must be > 0 • Update transaction balance_due • Update customer outstanding_balance • Audit log • For advance: transaction_id can be null

GET

/payments

Auth: JWT

Purpose: List all payment records.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?customer_id=uuid&mode;=UPI&page;=1</pre>	<pre>{ "items": [{ "id": "uuid", "amount": 5000, "mode": "UPI", "customer": "Ramesh", "date": "2026-06-25", ... }] }</pre>	• Pagination, filter by mode/customer/date • Tenant-scoped

GET

/payments/{id}

Auth: JWT

Purpose: Get payment details.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{ "id": "uuid", "amount": 5000, "mode": "UPI", "reference": "UTR123456", "customer_id": "uuid" }</pre>	• Ownership check

4.11

Voice Processing

Speech-to-transaction AI pipeline

POST

/voice/process

Auth: JWT

Purpose: Upload audio and create a transaction draft.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>multipart/form-data (MP3/WAV/OGG/M4A)</pre>	<pre>{ "draft_id": "uuid", "items": [{ "name": "Coke", "qty": 2, "price": 40 }], "confidence": 0.94 }</pre>	• Validate audio format and size • Speech-to-text transcription • Gemini entity extraction (customer, items, qty) • ProductMatchingService — alias + fuzzy match • Create draft in voice_transcripts • Return preview for confirmation

POST

/voice/confirm

Auth: JWT

Purpose: Confirm a voice draft and create the final transaction.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{ "draft_id": "uuid", "edits": [{ "item_idx": 0, "qty": 3 }] }</pre>	<pre>{ "transaction_id": "uuid", "invoice_number": "SRD/2026-27/002" }</pre>	• Validate draft belongs to user • Apply user edits if any • Validate stock availability • Fetch real prices from DB • Create final transaction • Update stock and customer balance

GET

/voice/history

Auth: JWT

Purpose: Get history of voice processing sessions.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?page=1&status;=CONFIRMED</pre>	<pre>{ "items": [...] }</pre>	• Pagination, filter by status • Tenant-scoped

DELETE

/voice/history/{id}

Auth: JWT

Purpose: Delete a voice draft record (failed/test drafts only).

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{ "success": true }</pre>	• Only FAILED or DRAFT status records can be deleted • Cannot delete CONFIRMED records • Soft delete only

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Expenses

Business expense tracking

POST

/expenses

Auth: JWT

Purpose: Create a new expense record.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"title":"Fuel","amount":500,"category":"TRANSPORT","payment_mode":"CASH","date":"2026-06-24"}</pre>	<pre>{"expense_id":"uuid"}</pre>	• Validate amount > 0 • Validate date • Validate category • Tenant-scoped

GET

/expenses

Auth: JWT

Purpose: List all expenses.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?page=1&category;=TRANSPORT&from;=2026-04-01</pre>	<pre>{"items":[...],"total":25,"total_amount":45000}</pre>	• Pagination, filter by category/date/payment_mode • Tenant-scoped

GET

/expenses/{id}

Auth: JWT

Purpose: Get expense details.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>—</pre>	<pre>{"id":"uuid","title":"Fuel","amount":500,"category":"TRANSPORT"}</pre>	• Ownership check

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Reports & Export

Financial reports, GST summaries and data export

GET

/reports/sales

Auth: JWT

Purpose: Sales report with totals, GST breakdown, top products.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?from=2026-04-01&to;=2026-06-30&group;_by=month</pre>	<pre>{"total_sales":540000,"total_gst":27000,"invoices":142,"items":[...]}</pre>	• Date range required • group_by: day, week, month • Tenant-scoped

GET

/reports/expenses

Auth: JWT

Purpose: Expense report grouped by category.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?from=2026-04-01&to;=2026-06-30</pre>	<pre>{"total_expenses":85000,"by_category":[{"category":"RENT","amount":30000},...]}</pre>	• Date range required • Tenant-scoped

GET

/reports/gst

Auth: JWT

Purpose: GST report: CGST, SGST, IGST summary for filing.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?month=6&year;=2026	<pre>{"taxable_amount":500000,"cgst":12500,"sgst":12500,"igst":0,"total_gst":25000,"by_hsn":[...]}</pre>	• Month + year required • Groups by HSN/SAC code • Used for GSTR-1 / GSTR-3B preparation

GET

/reports/profit-loss

Auth: JWT

Purpose: Profit & Loss: Sales minus Expenses minus Purchase Cost.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?from=2026-04-01&to;=2026-06-30	<pre>{"revenue":540000,"purchase_cost":340000,"expenses":85000,"gross_profit":200000,"net_profit":115000}</pre>	• Date range required • Revenue from transactions • Purchases from purchase_bills • Expenses from expenses collection

GET

/reports/export

Auth: JWT

Purpose: Export any report as CSV or Excel.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?type=sales&from;=2026-04-01&to;=2026-06-30&format;=csv	Binary CSV / XLSX file download	• Valid types: sales, expenses, gst, customers, products, inventory • Valid formats: csv, xlsx • Generate file and return as download • Large exports processed async — returns download URL

4.14

AI Assistant

RAG-powered business AI for natural language queries

POST

/ai/chat

Auth: JWT

Purpose: Ask the business AI assistant a natural language question.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>{"message":"Show unpaid invoices this month"}</pre>	<pre>{"answer":"You have 12 unpaid invoices this month worth ₹45,000.", "context":[...]}</pre>	• Permission check • RAG retrieval from tenant data only • LLM inference (Gemini) • Save to chat history • Audit log • Never access other tenant data

GET

/ai/history

Auth: JWT

Purpose: Retrieve previous AI conversations.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?page=1	<pre>{"items":[...]}</pre>	• Pagination • Tenant-scoped chat history

DELETE

/ai/history/{id}

Auth: JWT

Purpose: Delete a specific AI conversation thread.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"success":true}</pre>	• Ownership check • Soft delete only — anonymise rather than purge

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Analytics & Dashboard

Real-time business insights

GET	/dashboard	Auth: JWT
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Purpose: Dashboard summary for today.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"today_sales":12000,"pending_dues":5000,"low_stock_count":3,"top_product":"Coke"}</pre>	- Aggregate tenant-scoped data - Cache result for 5 minutes - Include low_stock_count

GET	/analytics/top-products	Auth: JWT
-----	-------------------------	-----------

Purpose: Best selling products by qty or revenue.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?period=30d&sort;_by=revenue&limit;=10</pre>	<pre>{"items":[{"name":"Coke","qty":200,"revenue":8000},...]}</pre>	- period: 7d, 30d, 90d, 1y - sort_by: qty or revenue - Tenant-scoped

GET	/analytics/top-customers	Auth: JWT
-----	--------------------------	-----------

Purpose: Top customers by revenue.

REQUEST	RESPONSE	VALIDATION & PROCESSING
<pre>?period=30d&limit;=10</pre>	<pre>{"items":[...]}</pre>	- Filter by date range - Tenant-scoped

GET	/analytics/monthly-sales	Auth: JWT
-----	--------------------------	-----------

Purpose: Monthly sales trend (12-month rolling).

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"months":[{"month":"Jan 2026","sales":45000,"invoices":38},...]}</pre>	- Return 12-month rolling data - Include invoice count per month - Tenant-scoped

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Files & Search

File storage and global full-text search

POST	/files/upload	Auth: JWT
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Purpose: Upload a file (invoice PDF, receipt image, audio).

REQUEST	RESPONSE	VALIDATION & PROCESSING
multipart/form-data	<pre>{"file_id":"uuid","url":"https://cdn/..."} </pre>	- Virus scan - File size validation (max 10MB) - Allowed: PDF, JPEG, PNG, MP3, WAV - Store metadata in DB - Upload to S3

GET

/files/{id}

Auth: JWT

Purpose: Download an uploaded file.

REQUEST	RESPONSE	VALIDATION & PROCESSING
–	Binary file response (streamed from S3)	• Authorization check • Stream from S3

DELETE

/files/{id}

Auth: JWT

Purpose: Delete an uploaded file.

REQUEST	RESPONSE	VALIDATION & PROCESSING
–	{"success":true}	• Soft delete metadata • Mark as deleted_at • S3 object retained for 30 days

GET

/search

Auth: JWT

Purpose: Global search across customers, invoices, products, suppliers.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?q=ravi&include;=customers,invoices,products	{"customers":[...],"invoices":[...], "products":[...], "suppliers":[...]}	• Search only tenant-scoped data • Full-text search via MongoDB text indexes • include param filters result types

4.17

Notifications & Audit

In-app notifications and audit trail

GET

/notifications

Auth: JWT

Purpose: Fetch all notifications (unread first).

REQUEST	RESPONSE	VALIDATION & PROCESSING
?page=1	{"items":[{"id":"uuid","message":"Low Stock: Coke","read":false,"priority":"HIGH"},...]}	• Unread first ordering • Tenant-scoped • TTL: auto-deleted after 90 days

PATCH

/notifications/{id}/read

Auth: JWT

Purpose: Mark a notification as read.

REQUEST	RESPONSE	VALIDATION & PROCESSING
–	{"success":true}	• Ownership check • Set read_at timestamp

GET

/audit-logs

Auth: JWT

Purpose: Get audit log history for this tenant.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?action=INVOICE_CREATED&from;=2026-06-01&page;=1	{"items":[{"action":"INVOICE_CREATED","user":"uuid","ts":"..."},...]}	• Tenant-specific only • Filter by action, user, date range • Pagination • TTL: auto-archived after 1 year

4.18

OCR — Future Feature

AI-powered invoice scanning and import

POST

/ocr/process-invoice

Auth: JWT

Purpose: Extract line items from a scanned invoice or PDF.

REQUEST	RESPONSE	VALIDATION & PROCESSING
multipart/form-data (PDF or image)	<pre>{"draft_id": "uuid", "items": [{"name": "Coke", "qty": 10, "rate": 40, "hsn": "2202", "gst": 12}, ...]}</pre>	• Validate file type (PDF/JPEG/PNG) • AI-powered OCR extraction • Match products against catalogue • Return structured draft for user review • Supports purchase bill import and customer invoice import

4.19

Admin

Platform-level administration — Admin JWT only

GET

/admin/users

Auth: Admin JWT

Purpose: List all platform users.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?page=1&role;=OWNER	<pre>{"items": [...]}</pre>	• Admin role required • Pagination, filter by role

GET

/admin/businesses

Auth: Admin JWT

Purpose: List all businesses on the platform.

REQUEST	RESPONSE	VALIDATION & PROCESSING
?page=1&status;=ACTIVE	<pre>{"items": [...]}</pre>	• Admin role required • Filter by status

GET

/admin/reports

Auth: Admin JWT

Purpose: Platform-level aggregate reports.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"users": 100, "businesses": 20, "invoices_today": 350, "revenue_today": 850000}</pre>	• Admin role required • Aggregate queries across all tenants

4.20

Health Check

Service liveness and dependency status

GET

/health

Auth: Public

Purpose: Service health check endpoint.

REQUEST	RESPONSE	VALIDATION & PROCESSING
—	<pre>{"status": "UP", "database": "CONNECTED", "redis": "CONNECTED", "ai": "AVAILABLE", "storage": "CONNECTED"}</pre>	• No authentication required • Used by load balancers, monitoring, Kubernetes liveness probes



Production Design Rules

Non-negotiable constraints for all endpoints

■ No Hard Deletes	Never delete Customers, Products, Users, Transactions, Suppliers. Use deleted_at / status / reversal entries.
■ API Versioning	Every endpoint under /api/v1. Future breaking changes move to /api/v2. Deprecation period minimum 6 months.
■ Pagination	All list endpoints must support page, size, sort, search. Default page size: 20. Max: 100.
■ JWT + RBAC	All non-public endpoints validated via JWT. Role enforcement: OWNER > STAFF > VIEWER. Admin endpoints require ADMIN role.
■ Audit Logging	Audit log for: invoice create/cancel/reverse, product price change, stock adjustment, user role change, settings update, login/logout.
■ Multi-Tenancy	Every query must include store_id filter. Leaking cross-tenant data is a critical security defect. Enforced at Service layer.
■ Soft Delete Pattern	Use deleted_at timestamp. Filter out in all list queries by default. Expose ?include_deleted=true for admin views only.
■ Rate Limiting	Public: OTP 5/hr, Login 10/hr. Authenticated: 1000 req/min per store. AI endpoints: 60/hr per store.
■ Backend Pricing	Backend fetches selling_price from products collection on every invoice. Client-submitted item prices are always ignored.
■ Stock Integrity	Validate stock before creating invoices. Restore stock on cancellation/reversal. Use atomic \$inc — never read-modify-write.
■ GST Logic	CGST+SGST for same state (store.state_code == supply_state). IGST for inter-state. Backend always determines tax type.
■ Feature Flags	Business Mode endpoints (Suppliers, Purchases, GST, Reports) must check store.enabled_features before processing.

Summary: 20 API modules • 70+ endpoints • Base: /api/v1 • New in v4.0: Suppliers, Purchases, Payments, Reports/Export, Categories, Feature Flags (/settings/features), DELETE /voice/history/{id}, DELETE /ai/history/{id}, GET /reports/gst, GET /reports/profit-loss